

Incidence and prognosis of cardiac conduction system diseases in hypertension: the STEP trial

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Supplementary Information

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STROBE checklist. STROBE Statement

Supplement Table 1. Baseline characteristics of the study participants without preexisting CSD stratified by the treatment groups.

Characteristics	Standard treatment n =3166	Intensive treatment n =3193	P Value
Age, years	65 (62, 69)	65 (62, 69)	0.237
Male sex, no. (%)	1389 (43.9)	1431 (44.8)	0.464
BMI, kg/m ²	25.59±3.18	25.52±3.13	0.464
Baseline blood pressure, mm Hg			
SBP	146.23±16.69	146.32±16.68	0.827
DBP	82.48±10.63	82.8±10.68	0.224
eGFR	109.20±23.75	110.31±24.04	0.064
Renal dysfunction, no. (%)			
eGFR < 60 ml/min/1.73 m ²	53 (1.5)	56 (1.4)	0.811
Fasting serum glucose, mmol/L	5.78 (5.27-6.63)	5.77 (5.25-6.57)	0.189
Lipids profile, mmol/L			
Total cholesterol	4.91±1.06	4.91±1.57	0.128
Triglycerides	1.35 (0.96-1.92)	1.33 (0.94-1.95)	0.737
HDL-C	1.27±0.31	1.26±0.31	0.301
Smoking history, no. (%)	481 (15.2)	498 (15.6)	0.705
Medical history, no. (%)			
Diabetes mellitus	601 (19.0)	620 (19.4)	0.683
Hyperlipidemia	1159 (36.6)	1190 (37.3)	0.603
Cardiovascular disease	198 (6.3)	197 (6.2)	0.931
Aspirin use, no. (%)	313 (9.9)	338 (10.6)	0.380
Statin use, no. (%)	612 (19.3)	615 (19.3)	0.969
ARB use, no. (%)	1380 (43.6)	1415 (44.3)	0.576
The 10-year Framingham risk ≥15%, no. (%)	2425 (77.0)	2460 (77.5)	0.660

Continuous variables represented as mean ± SD or median (IQR), and qualitative parameters are presented as number (%). Continuous variables were compared using the student t-test if normally distributed, or compared using the Wilcoxon Rank-sum test if non-normally distributed. Categorical variables were compared using the chi-square test. Two-sided $P < 0.05$ was defined as statistically significant; multiple comparisons were not adjusted.

Abbreviations: CSD, conduction system diseases; BMI, body mass index; SBP, systolic blood pressure; DBP, diastolic blood pressure; eGFR, estimated glomerular filtration rate; HDL-C, high-density lipoprotein cholesterol; ARB, angiotensin receptor blocker; SD, standard deviation; IQR, interquartile range.

Supplement Table 2. Type of new-onset CSD detected according to the treatment groups.

Type of new-onset CSD	Standard treatment (n=3166)	Intensive treatment (n=3193)
CSD	188	205
First-degree atrioventricular block	115	122
LVCD	31	36
LBBB	4	5
Left anterior branch block	20	21
Left posterior branch block	3	0
Nonspecific intraventricular conduction delay	4	10
RBBB	51	58

Abbreviations: CSD, conduction system diseases; LVCD, left ventricular conduction disease; LBBB, left bundle branch block; RBBB, right bundle branch block.

Supplement Table 3. Interaction between randomization assignment, age, and the risk of incident CSD ^a.

Variable	Age ≥ 66.2 years ^b	HR (95% CI)	P for interaction
CSD	Yes	1.17 (0.87, 1.59)	0.962
	No	0.96 (0.74, 1.26)	
First-degree atrioventricular block	Yes	1.13 (0.77, 1.66)	0.988
	No	0.98 (0.69, 1.38)	
LVCD	Yes	1.29 (0.66, 2.54)	0.922
	No	0.81 (0.39, 1.72)	
RBBB	Yes	1.38 (0.77, 2.48)	0.823
	No	0.96 (0.59, 1.57)	

^a The Fine–Gray subdistribution hazard model was used for analysis and adjusted for age, sex, BMI, baseline SBP, smoking history, history of hyperlipidemia, history of cardiovascular disease, aspirin use, statin use, ARB use, eGFR, total cholesterol, HDL-C. Two-sided $P < 0.05$ was defined as statistically significant; multiple comparisons were not adjusted.

^b Age is stratified by the mean age in study cohort.

Abbreviations: CSD, conduction system diseases; LVCD, left ventricular conduction disease; RBBB, right bundle branch block; HR, hazard ratio; CI, confidence interval; BMI, body mass index; SBP, systolic blood pressure; eGFR, estimated glomerular filtration rate; HDL-C, high-density lipoprotein cholesterol, ARB, angiotensin receptor blocker.

Supplement Table 4. Interaction between randomization assignment, diabetes mellitus, and the risk of incident CSD ^a.

Variable	Diabetes mellitus	HR (95% CI)	<i>P</i> for interaction
CSD	Yes	0.97 (0.64, 1.47)	0.777
	No	1.07 (0.86, 1.34)	
First-degree atrioventricular block	Yes	0.98 (0.59, 1.63)	0.951
	No	1.05 (0.78, 1.41)	
LVCD	Yes	1.06 (0.25, 4.49)	0.830
	No	1.14 (0.68, 1.91)	
RBBB	Yes	0.98 (0.40, 2.43)	0.928
	No	1.10 (0.73, 1.66)	

^a The Fine–Gray subdistribution hazard model was used for analysis and adjusted for age, sex, BMI, baseline SBP, smoking history, history of hyperlipidemia, history of cardiovascular disease, aspirin use, statin use, ARB use, eGFR, total cholesterol, HDL-C. Two-sided $P < 0.05$ was defined as statistically significant; multiple comparisons were not adjusted.

Abbreviations: CSD, conduction system diseases; LVCD, left ventricular conduction disease; RBBB, right bundle branch block; HR, hazard ratio; CI, confidence interval; BMI, body mass index; SBP, systolic blood pressure; eGFR, estimated glomerular filtration rate; HDL-C, high-density lipoprotein cholesterol, ARB, angiotensin receptor blocker.

Supplement Table 5. Risk of incident CSD or pacemaker in the intensive treatment group compared with the standard treatment group ^a.

Outcome	No. of events	Model ^b	HR (95% CI)	P value
New pacemaker	19	Unadjusted	0.58 (0.23, 1.47)	0.247
		Adjusted	0.61 (0.26, 1.43)	0.254
CSD or new pacemaker	403	Unadjusted	1.05 (0.87, 1.28)	0.597
		Adjusted	1.04 (0.86, 1.27)	0.667

^a The Fine–Gray subdistribution hazard model was used for analysis. Two-sided $P < 0.05$ was defined as statistically significant; multiple comparisons were not adjusted.

^b Adjusted model included age, sex, BMI, baseline SBP, smoking history, history of diabetes mellitus, history of hyperlipidemia, history of cardiovascular disease, aspirin use, statin use, ARB use, eGFR, total cholesterol, HDL-C. Abbreviations: CSD, conduction system diseases; HR, hazard ratio; CI, confidence interval; BMI, body mass index; SBP, systolic blood pressure; eGFR, estimated glomerular filtration rate; HDL-C, high-density lipoprotein cholesterol; ARB, angiotensin receptor blocker.

Supplement Table 6. Risk of incident CSD in the intensive treatment group compared with the standard treatment group using inverse probability weighting to account for missing follow-up data ^a.

Variable	Model ^b	HR (95% CI)	P value
CSD	Unadjusted	1.14 (0.90, 1.43)	0.273
	Adjusted	1.19 (0.95, 1.49)	0.130
First-degree atrioventricular block	Unadjusted	0.96 (0.72, 1.26)	0.758
	Adjusted	1.03 (0.79, 1.34)	0.849
LVCD	Unadjusted	0.96 (0.55, 1.67)	0.873
	Adjusted	1.09 (0.67, 1.76)	0.737
RBBB	Unadjusted	1.24 (0.75, 2.07)	0.399
	Adjusted	1.57 (0.97, 2.55)	0.067

^a The Fine–Gray subdistribution hazard model was used for analysis. Two-sided $P < 0.05$ was defined as statistically significant; multiple comparisons were not adjusted.

^b Adjusted model included age, sex, BMI, baseline SBP, smoking history, history of diabetes mellitus, history of hyperlipidemia, history of cardiovascular disease, aspirin use, statin use, ARB use, eGFR, total cholesterol, HDL-C. Abbreviations: CSD, conduction system diseases; LVCD, left ventricular conduction disease; RBBB, right bundle branch block; HR, hazard ratio; CI, confidence interval; BMI, body mass index; SBP, systolic blood pressure; eGFR, estimated glomerular filtration rate; HDL-C, high-density lipoprotein cholesterol, ARB, angiotensin receptor blocker.

Supplement Table 7. Adverse cardiovascular events and secondary outcomes comparing participants with new-onset CSD versus those free of CSD ^a.

Variable	No. of events	Model ^c	β	SE	HR (95% CI)	χ^2	P value
Adverse cardiovascular events ^b	213	Unadjusted	-0.099	0.297	0.91 (0.51, 4.62)	-0.333	0.739
		Adjusted	-0.178	0.301	0.84 (0.47, 1.50)	-0.598	0.550
Components of adverse cardiovascular events							
Stroke	74	Unadjusted	-0.445	0.589	0.64 (0.20, 2.03)	-0.757	0.449
		Adjusted	-0.422	0.593	0.66 (0.21, 2.05)	-0.726	0.468
Acute coronary syndrome	92	Unadjusted	0.234	0.393	1.26 (0.58, 2.73)	0.594	0.552
		Adjusted	0.129	0.402	1.14 (0.50, 2.58)	0.309	0.758
Acute decompensated heart failure	3	Unadjusted	-	-	-	-	-
		Adjusted	-	-	-	-	-
Coronary revascularization	36	Unadjusted	0.654	0.530	1.92 (0.68, 5.42)	1.235	0.217
		Adjusted	0.676	0.549	1.97 (0.66, 5.88)	1.209	0.227
Atrial fibrillation	27	Unadjusted	-	-	-	-	-
		Adjusted	-	-	-	-	-
Death from cardiovascular causes	16	Unadjusted	0.014	1.033	1.02 (0.13, 7.68)	0.014	0.989
		Adjusted	-0.306	1.075	0.74 (0.19, 2.82)	-0.446	0.656
Death from any cause	49	Unadjusted	0.587	0.473	1.80 (0.71, 4.54)	1.242	0.214
		Adjusted	0.390	0.483	1.48 (0.57, 3.81)	0.807	0.420

^a In the analyses of the adverse cardiovascular events and secondary outcomes except for death from any cause, the HR, 95% CI, and P value were calculated using of the Fine–Gray subdistribution hazard model for the competing risk of death. Cox regression model was used for death from any cause. New-onset CSD was cast as a time-varying covariate. Two-sided $P < 0.05$ was defined as statistically significant; multiple comparisons were not adjusted.

^b The adverse cardiovascular events were a composite of stroke, acute coronary syndrome, acute decompensated heart failure, coronary revascularization, atrial fibrillation, or death from cardiovascular causes.

^c Risk of acute decompensated heart failure and atrial fibrillation were not shown due to insufficient number of events.

^d Adjusted models included age, sex, BMI, baseline SBP, smoking history, history of diabetes mellitus, history of hyperlipidemia, history of cardiovascular disease, aspirin use, statin use, ARB use, eGFR, total cholesterol, HDL-C and treatment strategy.

Abbreviations: CSD, conduction system diseases; HR, hazard ratio; CI, confidence interval; BMI, body mass index; SBP, systolic blood pressure; eGFR, estimated glomerular filtration rate; HDL-C, high-density lipoprotein cholesterol, ARB, angiotensin receptor blocker

STROBE Statement—Checklist of items that should be included in reports of *cohort studies*

	Item No	Recommendation	Page No
Title and abstract	1	(a) Indicate the study's design with a commonly used term in the title or the abstract	2
		(b) Provide in the abstract an informative and balanced summary of what was done and what was found	2
Introduction			
Background/rationale	2	Explain the scientific background and rationale for the investigation being reported	2-3
Objectives	3	State specific objectives, including any prespecified hypotheses	3
Methods			
Study design	4	Present key elements of study design early in the paper	12-13
Setting	5	Describe the setting, locations, and relevant dates, including periods of recruitment, exposure, follow-up, and data collection	13-14
Participants	6	(a) Give the eligibility criteria, and the sources and methods of selection of participants. Describe methods of follow-up (b) For matched studies, give matching criteria and number of exposed and unexposed	13-14 n/a
Variables	7	Clearly define all outcomes, exposures, predictors, potential confounders, and effect modifiers. Give diagnostic criteria, if applicable	12-15
Data sources/ measurement	8*	For each variable of interest, give sources of data and details of methods of assessment (measurement). Describe comparability of assessment methods if there is more than one group	14-15
Bias	9	Describe any efforts to address potential sources of bias	15-17
Study size	10	Explain how the study size was arrived at	13,15
Quantitative variables	11	Explain how quantitative variables were handled in the analyses. If applicable, describe which groupings were chosen and why	15-16
Statistical methods	12	(a) Describe all statistical methods, including those used to control for confounding (b) Describe any methods used to examine subgroups and interactions (c) Explain how missing data were addressed (d) If applicable, explain how loss to follow-up was	15-17 16 16 16

		addressed (e) Describe any sensitivity analyses	16
Results			
Participants	13*	(a) Report numbers of individuals at each stage of study—eg numbers potentially eligible, examined for eligibility, confirmed eligible, included in the study, completing follow-up, and analysed (b) Give reasons for non-participation at each stage (c) Consider use of a flow diagram	3-4 4 4
Descriptive data	14*	(a) Give characteristics of study participants (eg demographic, clinical, social) and information on exposures and potential confounders (b) Indicate number of participants with missing data for each variable of interest (c) Summarise follow-up time (eg, average and total amount)	3-4 4 4
Outcome data	15*	Report numbers of outcome events or summary measures over time	4-5
Main results	16	(a) Give unadjusted estimates and, if applicable, confounder-adjusted estimates and their precision (eg, 95% confidence interval). Make clear which confounders were adjusted for and why they were included (b) Report category boundaries when continuous variables were categorized (c) If relevant, consider translating estimates of relative risk into absolute risk for a meaningful time period	4-6 n/a n/a
Other analyses	17	Report other analyses done—eg analyses of subgroups and interactions, and sensitivity analyses	4-5
Discussion			
Key results	18	Summarise key results with reference to study objectives	6
Limitations	19	Discuss limitations of the study, taking into account sources of potential bias or imprecision. Discuss both direction and magnitude of any potential bias	11-12
Interpretation	20	Give a cautious overall interpretation of results considering objectives, limitations, multiplicity of analyses, results from similar studies, and other relevant evidence	11-12
Generalisability	21	Discuss the generalisability (external validity) of the study results	11-12
Other information			
Funding	22	Give the source of funding and the role of the funders for the present study and, if applicable, for the original study on which the present article is based	18

*Give information separately for exposed and unexposed groups.

Note: An Explanation and Elaboration article discusses each checklist item and gives methodological background and published examples of transparent reporting. The STROBE checklist is best used in conjunction with this article (freely available on the Web sites of PLoS Medicine at <http://www.plosmedicine.org/>, Annals of Internal Medicine at <http://www.annals.org/>, and Epidemiology at <http://www.epidem.com/>). Information on the STROBE Initiative is available at <http://www.strobe-statement.org>.